Mock Regional

11 problems, 5 hours, 1 computer — then the debrief

Learning goals

- Run a full regional simulation under real rules
- Practise triage, the problem board, and keyboard scheduling
- Recognise every technique from Lectures 2–9 in disguise
- Debrief systematically: what cost time, what cost penalty, what to train next

Rules for the mock

- 5 hours, teams of 3, **one computer**, printed notebook allowed
- Standard scoring: solved, then penalty (20 min per rejected try)
- Scoreboard frozen for the last hour
- No internet except the judge; clarifications via the coach
- Problems in alphabetical order — **not** sorted by difficulty

The set at a glance (no spoilers)

	Title		Title
A	Odometer Without Fours	G	Palindrome Banner
B	Arrow Grid	H	Rest Stops
C	Toll Vouchers	I	Coprime Committees
D	Lab Bookings	J	Fence Lattice
E	Double Coverage	K	Forbidden Jingle
F	Gears		

- Statements are summarised in the notes; hand them out printed

Recommended first 20 minutes

- Split: A–D, E–H, I–K (+ template typing)
- Label E / M / H with one-line ideas
- First keyboard: whichever **E** is found first
- Expectation: 2 easy, 1 easy-medium, 5 medium, 2 medium-hard, 1 hard

Debrief

Process, scoreboard, upsolving

Intended difficulty

	Difficulty
A	Easy
E	Easy
B	Easy-medium
C	Medium
D	Medium
F	Medium
G	Medium
H	Medium
J	Medium

Solutions

> **Solutions are not published here.** The coach holds the solutions and releases them only after the contest has been run.

- Problem-by-problem solutions are presented by the coach after the contest
- Before that: every team fills in its timeline (debrief template)

Running the judge and the log

- Any local judge: DOMjudge, a Codeforces Gym mashup, or the coach checking outputs by script
- Each team keeps the problem board from Lecture 1
- The board records when each problem was read, labelled, started and accepted
- The board is the main input to the debrief

Generic traps worth a second look

- Overflow: beyond $2 \cdot 10^9$ → `long long`; beyond 10^{18} → `__int128`
- Special outputs ("never", unreachable, stays forever) are part of the problem
- "At most" vs "exactly": reread every quantifier
- Modular subtraction without `+ MOD`
- Team level: nobody reading the last problems until the final hour

Expected scoreboard

Solved	Typical team
3	A, E, B — fundamentals in place
5–6	+ C, D or G, J — solid regional team
8–9	+ F, H, I — medal contention
10–11	+ K — regional winner

Debrief template (fill in as a team)

	Read at	Label	Started	AC at	Tries	Time lost on	Root cause
A							
…							

- Root cause ∈ {misread, wrong idea, implementation bug, overflow, TLE, format, idle machine}

Debrief questions

1. Which problem did we solve **latest relative to the field**? Why?
2. How long was the machine idle or used for thinking?
3. Which WA could the 60-second checklist have prevented?
4. Which unsolved problem was closest? What was missing — idea or code?
5. What goes into the notebook / training plan this week?

Upsolving rule

- Every problem: re-solve alone within one week
- Unsolved medium: write the solution, then **explain it** to a teammate
- Add a stress test for every WA you got
- Update the problem board template with what you learned

Summary

- The mock exercises every lecture of this track under contest pressure
- Most points are in the medium band: C, D, F, G, H, J
- The debrief matters more than the score
- Next: National Championship Prep — training systems, advanced data structures, DP optimisations